AOUANOUK Slimane

+33 7 83 54 30 26 — aouanoukslim@gmail.com — LinkedIn — slimaneaouanouk.com

Available for a 6-month internship starting February 2027 in robotics or autonomous systems.

Graduate Engineering student at Arts et Métiers pursuing a **Research Master's (M2) in Advanced Systems and Robotics at Sorbonne Université**. Research experience in **trajectory optimization, perception, real-time control, and physical validation**. Currently developing a **Vision-Language-Action pipeline for a Unitree G1 humanoid**.

EDUCATION — Combined GPA : 3.92 / 4.00

Sorbonne Université — Paris, France **Sept 2026 – Aug 2027**

Research Master's (M2) in Advanced Systems & Robotics (SAR)

Selected research-oriented coursework: Mobile Robotics (ROS 2, SLAM), Robot Perception (camera-based vision and multi-sensor fundamentals), Machine Learning (deep learning, CNNs/RNNs, autoencoders, MDPs, reinforcement learning, DQN/DDPG, PyTorch), State Estimation & System Identification (Bayesian estimation, EKF, neural extended Kalman filter (NEKF), UKF, particle filters, and data-driven identification), Nonlinear Control, and Robot Manipulators.

Arts et Métiers Institute of Technology (ENSAM) — Paris, France **Sept 2024 – Jul 2027**

Combined BS × MS Engineering Degree — *Member of **ParisTech**; rigorous multidisciplinary training supported by **world-class research laboratories**.*

- **Ranked top 12% of cohort** (**142 / 1223** students)
- **Relevant Coursework:** Solid Mechanics, Robotics and Mechatronics, Numerical Optimization, Computer-Aided Design (CAD), Experimental Mechanics and Validation Testing, Fluid Mechanics

WORK EXPERIENCE

Trajectory Optimization and Collision Avoidance for Autonomous Systems **Jan–Aug 2026**

Robotics Research Project & Internship — PIMM Lab / CNRS@CREATE; Paris (Jan–May) → Singapore (Jun–Aug) — Supervisors: Prof. Amine Ammar, Prof. Francisco Chinesta and Prof. Dominique Baillargeat

- **Co-developed a Python/OpenCV visual-localization pipeline** using a Basler overhead camera and ArUco markers to estimate rover position and heading for closed-loop mobile-robot navigation.
- **Calibrated camera intrinsics using 20 ChArUco views**, achieving **0.191 px reprojection RMS**; combined lens-distortion correction with metric homography to reduce mean localization error **84%, from 36.3 to 5.8 mm**, on an independent nine-point ground grid.
- **Diagnosed and corrected height-induced parallax**, reducing mean rover-marker localization error from **60.8 to 11.6 mm** across five physical validation positions.
- **Profiled and optimized vision processing through image downsampling and point-wise distortion correction**, integrating localization, trajectory correction, serial rover control, and experiment logging at a median **19.84–19.86 Hz**.
- **Validated the integrated navigation stack through two frozen 20-run physical campaigns**, achieving **40/40 goal arrivals with zero collisions** and collecting synchronized control and trajectory logs.
- **Evaluated trajectory-planning and control configurations through a 100,800-case simulation campaign** and quantified simulation-to-real discrepancies across **39 paired physical and digital-twin runs**.
- **Co-developed pytest unit and regression tests** covering calibration, homography, controller behavior, and analytical gradients, including finite-difference checks for numerical consistency.
- **First author of a robotics manuscript currently under journal review**, presenting the trajectory-planning framework, experimental validation, and simulation-to-real evaluation.

SELECTED PROJECTS

Humanoid Robot Learning for Assistive Manipulation — VLA **2026–27**

Sorbonne Université · Supervised by Prof. Olivier Bruneau, LURPA / ENS Paris-Saclay

- **Developing an end-to-end Vision-Language-Action (VLA) pipeline** for autonomous manipulation on a Unitree G1 humanoid, from teleoperated demonstrations to real-world deployment.
- **Targeting OpenPI (π_0/$\pi_{0.5}$) policy fine-tuning** using imitation learning and multimodal datasets combining natural-language instructions, RGB/RGB-D observations, and robot actions.
- **Exploring language-conditioned manipulation policies** for household assistance tasks, targeting robust execution, generalization to unseen configurations, and quantitative real-world evaluation.
- **Exploring AI-based motion optimization in MuJoCo or NVIDIA Isaac Lab (Isaac Sim)**, with motion capture and human-to-robot retargeting envisaged to adapt human movements to the robot.

AlSi7 Bearing Support: Casting, Machining & Metrology – Arts et Métiers **2025**

- Made the sand mold, cast the AlSi7 part and machined it on a conventional lathe and CNC mill (**2 phases**).
- Inspected **6 dimensional and geometric characteristics** using a CMM; measured **0.015 mm flatness against a 0.100 mm limit**, investigated nonconformities and proposed mold and setup corrections.

SKILLS AND INTERESTS

Programming & ML: Python (data analysis, numerical computation, scientific optimization), PyTorch (current coursework), MATLAB, Linux (current coursework), LaTeX

Robotics & Vision: Multi-robot hardware integration, ArUco localization, camera calibration, real-time trajectory testing, ROS 2 (current coursework)

Modeling & Simulation: CFD (STAR-CCM+), FEA (Abaqus), CAD (Fusion 360, 3DEXPERIENCE), Simulink, digital twins

Languages: French (native), English (TOEFL iBT: 5/6), Italian (basic), Spanish (basic)

Interests: Robot learning, physical AI, applied reinforcement learning, basketball, strength training